

Answer Lab: Say It, Support It, Improve It

GROW • Science • 40 minutes

I can give a scientific answer with evidence and reasoning, then improve it after audience response.

Name: _____ Date: _____

You may write, type, say, sign, point or direct a partner. An adult can read the words and record your exact response. Keep the choice and explanation your own.

Words for this lesson

Word	Meaning
claim	my answer
reasoning	why the evidence matters
revise	make meaning clearer

First idea

Which addition would most improve: 'The rover went farther on smooth surface'?

- A. Add the repeated distances and explain friction
- B. Add three unrelated space facts
- C. Use bigger lettering

My first idea and reason:

CLAIM

The rover travels farther on the smooth surface.

Claim

We do • choose, explain, check

Which revision is scientifically strongest?

A. The rover went farther on smooth surface: 44–46 cm versus 18–20 cm, because fabric produced more friction.

B. The rover was excellent, amazing, fast and very interesting on the table.

C. The Earth rotates once in about 24 hours, and the Moon reflects sunlight.

My choice: _____ Evidence or reason:

Three-Topic Answer Lab

Match each complete question–evidence–reason answer or honest limitation to Forces, Space or Climate. Freeze one answer and its limit, then improve it after an audience check.

Read or hear one card. Choose a group. Give the evidence. A partner checks your reason before you move on. Point or write card numbers; cutting is optional.

Group	Card numbers and evidence
FORCES	
SPACE	
CLIMATE	

Activity cards

1 • FORCES ANSWER • Why less far on fabric? Practice evidence: smooth 46, 44, 45 cm; fabric 18, 20, 19 cm. Greater fabric friction opposed motion more.	2 • FORCES LIMIT • one small rover One rover and two classroom surfaces cannot represent every vehicle or terrain.
3 • SPACE ANSWER • Why day and night? Model evidence: a marked place turns into and out of the lamp's light. Earth rotates, so day and night repeat.	4 • SPACE LIMIT • simplified lamp and globe The model is not to scale and fixing the lamp is an assumption; it shows how rotation explains the pattern.

5 • CLIMATE ANSWER • Why can extra gas warm Earth?

Energy-route evidence: more outgoing infrared is absorbed and some returns towards the surface, slowing net energy loss and raising the global average.

6 • CLIMATE LIMIT • arrows simplify a complex system

The energy route shows direction, not real amounts, times or every climate feedback.

Read the complete answer and limit. What did the audience hear, and what will the pupil clarify, add or keep?

Your lesson record

Choose one embedded question—rover friction, day and night, or extra greenhouse gas and warming—give a first answer, have it received by an adult, then revise or justify keeping it.

Model helps us see

The visible first-and-next answer makes Voice, Audience and Influence real without turning the task into a score.

Model does not show

One revised answer shows today's reasoning, not everything the pupil knows. Spoken, pointed, drawn and exactly scribed routes are equivalent.

Supported

Choose one embedded topic answer, identify its claim, evidence and reason, then communicate it in a preferred mode.

Adult reads back exact choices; pupil selects Add evidence, Clarify reason or Keep and explain.

Standard

Answer one of the three embedded questions independently, receive one audience check and revise one part.

Use Claim, Evidence and Reason headings only if needed; fade them after the first success.

Stretch

Answer a second embedded topic question, compare evidence strength and justify the revision or no-change decision.

Add one limitation without weakening a secure conclusion.

Evidence target

First answer, adult read-back, pupil-selected next move and improved answer or justified decision to keep it.

Exit, Audience and evidence • W12B

Supported: Point to what changed between your first and next answer.

Standard: Explain how audience response made your Science clearer.

Stretch: Justify a change or no-change decision using relevance, evidence and limitation.

What I said, and what it changed

Next I will _____.

Adult witness note

What was observed, and with what level of independence?

My evidence and explanation

What does this evidence not show?

Visual reference cards

These are diagrams or model views from the uploaded lesson. Use their labels and the card descriptions; they are not photographs of your own results.

EVIDENCE

smooth 46, 44, 45 cm · fabric 18, 20, 19 cm
every fabric result is lower

Evidence

REASON

fabric creates more friction, opposing motion,
so the rover slows and stops sooner

Reason

Exit • one next step

After the adult has listened, what will you keep, improve or check next?
